Jurassic Polar Movement Relative to North America[1]

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Previous paleomagnetic studies of Jurassic rocks have not given concordant results and have led to the conclusion that the Jurassic pole position was possibly close to the present geographic pole. To test that supposition, the Kayenta, Carmel, Entrada, and Summerville formations were sampled from the extensive Jurassic sedimentary sequence in eastern Utah. The Lower and Upper Jurassic Kayenta and Summerville formations gave very similar magnetization directions. However, the Middle to Upper and Upper Jurassic Carmel and Entrada formations proved to be paleomagnetically unstable, and no reliable pole positions could be calculated. Poles calculated from two localities in the Kayenta formation and from the Summerville formation are 98°E, 59°N and 84°E, 59°N (Lower Jurassic) and 86°E, 70°N (Upper Jurassic, Oxfordian); thus they show a lack of major movement of the North American continent before Upper Jurassic (upper Oxfordian) time. Comparison with published Lower Cretaceous pole positions (135–100 m.y. in age) indicates that the 30 degrees of arc movement in the North American polar-wander curve (from the Upper Jurassic to the Lower Cretaceous pole positions) occurred in approximately 30 m.y. Thus the major portion of the spreading in the North Atlantic between Africa and North America did not occur until uppermost Jurassic-Lower Cretaceous times, making this period one of rapid sea-floor spreading compared to that of the Late Cenozoic.

The position of the paleomagnetic pole with respect to North America is fairly well established for the Carboniferous, Permian, Triassic, Cretaceous and Tertiary periods. Few satisfactory pole positions have been determined for the Jurassic period, and several of the poles that have been determined are near the position of the present axial dipole. The Jurassic, a period of about 60 million years [*Howarth*, 1964], is preceded and succeeded by paleomagnetically well-documented geologic periods; thus the lack of consistent paleomagnetic results is rather mystifying.

The importance of the Jurassic period lies in the large amount of polar wandering related to continental movement that occurs between Triassic and Cretaceous time (see Figure 1). Assuming the simplest path of movement, the change in pole position is of the order of 35°. If the magnetization direction acquired during the Jurassic was indeed the same as that of the axial dipole [*Opdyke and Wensink*, 1966; *Collinson and Runcorn*, 1960], the movement between Triassic and Cretaceous times was approximately 50°.

The seeming complexity of having the pole (continent) move from the Triassic position to its present latitude and then away (to the Cretaceous position) and back again (to the present position), coupled with the large arc distances involved per unit of time, has prompted the present study of Jurassic age rocks. It is the purpose of this study to begin to develop a Jurassic polar-wander path based on a series of stratigraphic sections of sediments varying in age from Upper Triassic to Lower Cretaceous.

PREVIOUS WORK

Collinson and Runcorn [1960] sampled the Kayenta and Carmel formations of Arizona and Utah. Their results gave pole positions indistinguishable from the position of the present axial dipole. *Opdyke and Wensink* [1966] studied the radiometrically dated Jurassic White Mountain series in New Hampshire and Vermont and got a similar result. One granite in the series was dated at 180 m.y. by K-Ar, Rb-Sr, and U-Pb techniques. However, it is noteworthy that all normal results from their mixed polarity study gave a present pole position but that the

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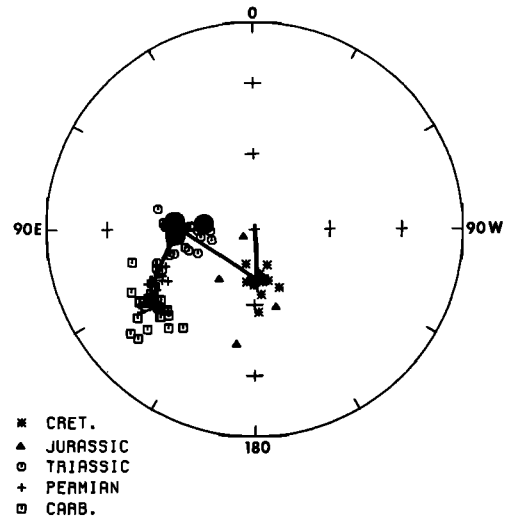


Fig. 1. Upper Paleozoic and Mesozoic published pole positions from North America. Poles are selected on the basis of the stability criteria given by *Irving* [1964] from data listed in the summaries by *Irving* [1964] and *McElhinney* [1968*a*, 1968*b*, 1969]. Heavy line indicates approximate polar-wander curve for this time.

reversed polarity results gave poles that formed a streak between the Cretaceous pole position and a pole off the northwest coast of Siberia (approximately 60°E, 80°N). Further, there seems to be an association of the normal directions with the monzanites of the series, whereas the reversed directions were observed in the gabbros. Either there was more than one time of igneous activity, or intrusion and cooling involved a long period of time, or the monzanites were not stably magnetized.

De Boer [1967] obtained a pole position from dikes cutting Upper Triassic and older formations and structures in the Appalachians. The mean pole position obtained from these dikes was midway between the Triassic and Cretaceous, and thus De Boer concluded that the dikes might be Jurassic in age. *Symons* [1969] has sampled rocks in British Columbia that are paleontologically indicated to be mid-Jurassic to Upper Cretaceous in age. Symons's results showed a pole close to that of the present dipole position and a pole some distance to the southwest of the North American Cretaceous poles. Since the area of sampling has been tectonically active since Jurassic time, the significance of these directions is uncertain.

Grommé et al. [1967] sampled plutonic rocks in the Sierra Nevada that gave radiometric ages in the Upper Jurassic (142 to 129 m.y.); however, the magnetic directions of the rocks were more closely associated with Cretaceous directions. *Grommé and Gluskoter* [1965] determined a pole position for the Jurassic Franciscan formation, which was situated in the North Atlantic Ocean quite removed from the North American polar-wander curve. The very anomalous position of this pole led Grommé and Gluskoter to conclude that either tectonic rotation of the Franciscan strata had occurred or an anomalous field existed at the time.

Geologic Setting

An extensive predominately nonmarine Jurassic sequence crops out throughout the Colorado plateau region, in Utah, Arizona, New Mexico, and western Colorado. In Utah the Jurassic formations are primarily red and buff-colored siltstone and sandstone. The sequence consists of seven formations shown in Figure 2. Samples were taken in the Kayenta, Carmel, Entrada, and Summerville formations at the approximate positions indicated by the arrows in Figure 2.

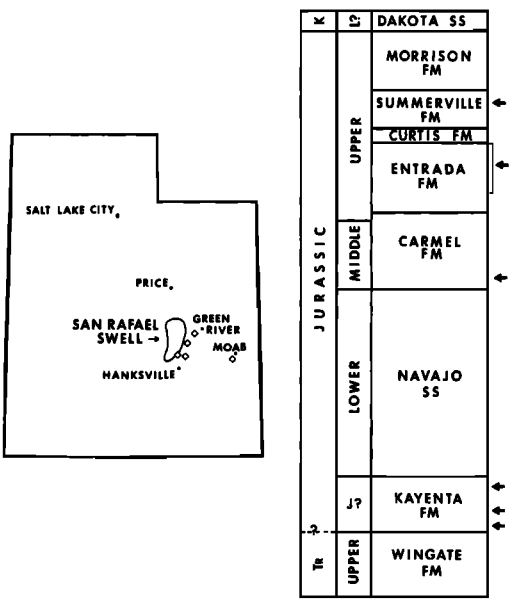


Fig. 2. Location map and the Jurassic stratigraphic sequence of Utah. Arrows indicate approximate position of sampled sections. Locations of the sampling sites are indicated by open diamonds on the map of Utah.

COLLECTION AND PREPARATION

All samples were collected by means of a hand-held gasoline-powered core drill and were oriented with a magnetic compass. All collections represent continuous stratigraphic sequences of the portions of the formation sampled. Cores were cut into 2.5-cm cylinders in the laboratory and measured on a PAR SM-1 spinner magnetometer. Stability tests were carried out by means of thermal demagnetization in a conventional noninductively wound furnace housed inside a three dimensional coil system; the entire apparatus was inside a 10-foot molypermalloy-shielded room. After each heating, the samples were cooled in a field of less than ± 10 γ and were again measured on the PAR magnetometer.

KAYENTA FORMATION

The Kayenta formation south of the sampling area has been described by *Harshbarger et al.* [1957]. The formation is characterized by reddish-purple to reddish-brown and buff fine-grained sandstone with subordinate interbedded mudstone and shale. The Kayenta formation is thought to span the Triassic-Jurassic boundary, since Triassic and Jurassic fossils (Jurassic and Triassic nonmarine mollusks and Early to Middle Jurassic and Late Triassic vertebrate remains) are found associated with each other in the formation. These associations indicate that the period boundary may be within or at the lower boundary of the Kayenta formation [*Harshbarger et al.*, 1957].

The formation was sampled south of Moab, Utah, in Kane Springs canyon (38.5°N, 190.6°W). At this locality the Kayenta consists of about 100 meters of alternating medium red to reddish-orange sandstone and dark red siltstone. Three short stratigraphic sequences separated from one another by 17 and 40 meters were sampled. The lower sequence, sampled from the Kayenta-Wingate boundary to 27 meters above the boundary, consisted of 54 samples; 17 meters higher, 2 meters of maroon siltstone and mudstone were sampled (14 samples). The third sequence (spanning 10 meters) sampled the uppermost beds of the formation and extended into the basal Navajo sandstone (26 samples).

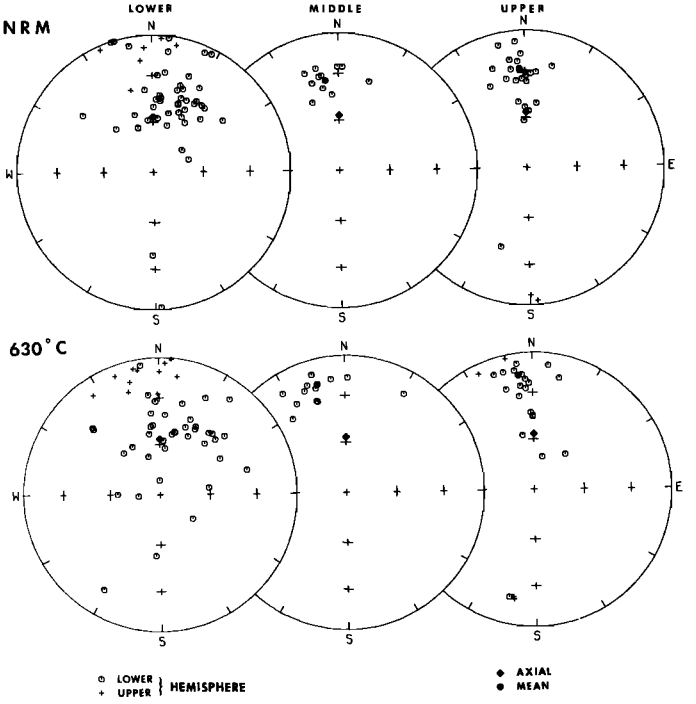


Fig. 3. NRM and thermal demagnetization directions from three sample sequences from the Kayenta formation. The direction of the axial dipole field is indicated by a diamond.

TABLE 1. Magnetization Directions and Pole Positions

Sampled Unit	Treatment[d]	Declination, deg	Inclination, deg	Number of Samples	Precision, k	α_{95}[e]	Pole Long.,[a] deg	Pole Lat.,[a] deg	dp[f]	dm[f]
Kayenta										
Lower										
All[b]	NRM	10.6	38.2	53	5.4	9.3				
All[b]	550	9.5	37.3	52	4.0	11.4				
All[b]	630	4.8	39.8	52	3.8	11.7				
Bottom	630	356.2	47.0	16	10.6	11.9	88.5	79.2	9.9	15.3
Center	630	46.1	50.1	13	17.6	10.2	22.4W	51.7	9.1	13.6
Top	630	354.2	-10.6	11	37.7	7.5	78.7	45.8	3.9	7.6
Middle										
	NRM	352.2	33.0	11	44.2	6.9	90.8	68.4	4.5	7.9
	550	347.9	19.1	11	21.5	10.1	94.2	59.3	5.5	10.5
	630	345.8	19.2	11	19.7	10.5	98.1	58.7	5.7	11.0
Upper										
	NRM	356.9	28.4	17	42.2	5.6	77.8	66.5	3.3	6.1
	550	354.5	14.8	15	24.1	7.9	80.8	58.7	4.2	8.2
	630	352.7	15.9	15	32.8	6.8	84.5	58.9	3.6	7.0
Carmel										
	NRM	357.6	50.5	47	15.4	5.5				
	500	357.7	48.7	42	9.9	7.4				
Entrada										
Dipping	NRM	8.5	60.3	57	66.4	2.3				
Flat	NRM	12.2	60.1	13	85.3	4.5				
Summerville										
All[b]	NRM	11.1	69.4	52	3.9	11.7	84.0W	74.8	17.0	19.9
All[b]	550	348.7	35.9	52	4.6	10.4	99.1	67.8	7.0	12.1
Sel[c]	550	354.1	37.0	40	10.2	7.5	86.3	70.1	5.1	8.7
All normal	550	350.2	34.9	12	4.7	22.5	94.9	67.7	14.9	25.9
All reversed	550	174.6	-39.1	38	2.6	18.2	85.8	71.7	13.0	21.7
Sel normal[c]	550	351.3	34.2	10	18.6	11.5	92.0	67.6	7.5	13.2
Sel reversed[c]	550	175.2	-38.0	30	8.7	9.4	83.8	71.0	66.6	11.2

[a]Longitude is east unless specified; latitude is north unless specified.
[b]All samples.
[c]Selected samples.
[d]Thermal demagnetization to 550°C or 630°C.
[e]α_{95}: Semiangle of 95% confidence cone.
[f]dp, dm: Semiaxis of error ellipse.

The natural remanent magnetization (NRM) directions of two of the three sequences in the Kayenta formation (middle and upper) are oriented approximately due north and are inclined about 33° downward. The third (lower sequence) has a similar direction but is quite scattered. Schmidt equal-area projection plots of the NRM directions for each locality are shown in Figure 3. Directions from all three sequences are of normal polarity, with the exception of three samples that appear to have reversed polarity. Two of the samples are at the base of the upper sequence section, and one is at the top of the upper sequence. The middle and upper sequences have well-grouped directions, whereas the lower sequence shows three different groups of directions. The NRM mean directions of the three sequences are within 15° of each other (see Table 1) and are removed from the present direction and the direction of the axial geocentric dipole field.

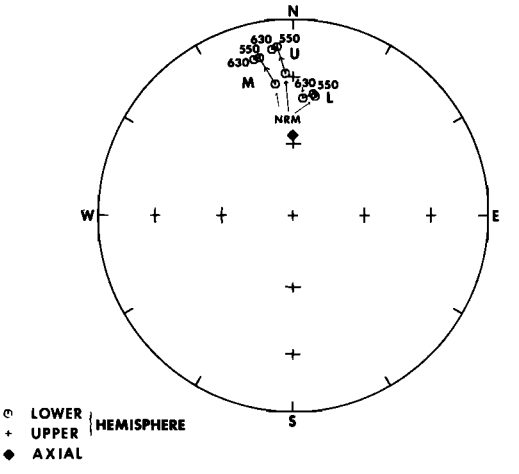


Fig. 4. Mean directions of the three sampled sequences before and after 550°C and 630°C demagnetization. Lower, middle, and upper sequences are designated by L, M, and U.

Thermal demagnetization experiments were carried out on all samples from each sequence. Samples were heated in two steps, to about 550°C and to approximately 630°C on the basis of experience with the red sediments of the Moenkopi formation [*Helsley*, 1969; *Helsley and Steiner*, 1972], which outwardly appear to be similar to the Kayenta sediments.

Figure 4 shows the progressive changes in the mean directions on thermal demagnetization of each of the sequences sampled. Heating the samples from the upper sequence to 550°C produced significant changes in direction of magnetization of a majority of the samples, notably a shallowing of inclinations by about 14°. The over-all cluster of these directions after 550° was decreased from that of the NRM directions. Heating of the samples to about 630°C caused slight systematic changes in individual directions and produced a slightly better clustering of the group (α_{95} 7.9° versus α_{95} 6.8°).

The upper sequence really contains three different directions. Stratigraphically, the top six samples have NRM directions near the direction of the axial dipole field, except for one sample of reverse direction. These directions remained unchanged on demagnetization. The seventeen samples below these six had uniform NRM directions that moved as a group on demagnetization and constitute 'the majority' referred to above. The basal three samples consist of two reversed directions and one near the present field direction. The one sample at the top and the two basal samples with reversed NRM directions moved toward a direction antipodal to that of the majority of samples on demagnetization. These results may indicate the presence of at least one, and probably two, reversals in the Lower Jurassic. Clearly, more sampling is necessary to substantiate these reversals, since both occurrences of reversed samples were associated with unstable samples.

A pole calculated for the upper sequence is shown in Figure 7 and Table 1. In this pole calculation the samples with directions near that of the axial dipole field were neglected. The samples with reversed directions were also ignored, since they did not attain completely antipodal directions on demagnetization and therefore may not be entirely free of secondary component.

Thermal demagnetization of the middle sequence to 550°C also produced a large shallowing of inclinations (about 15°) and increased the dispersion of the magnetization directions. Demagnetization of this group to about 630°C (Figure 3) again systematically changed the individual directions in the same direction as was observed in the upper sequence. The middle-sequence directions are well grouped, and the

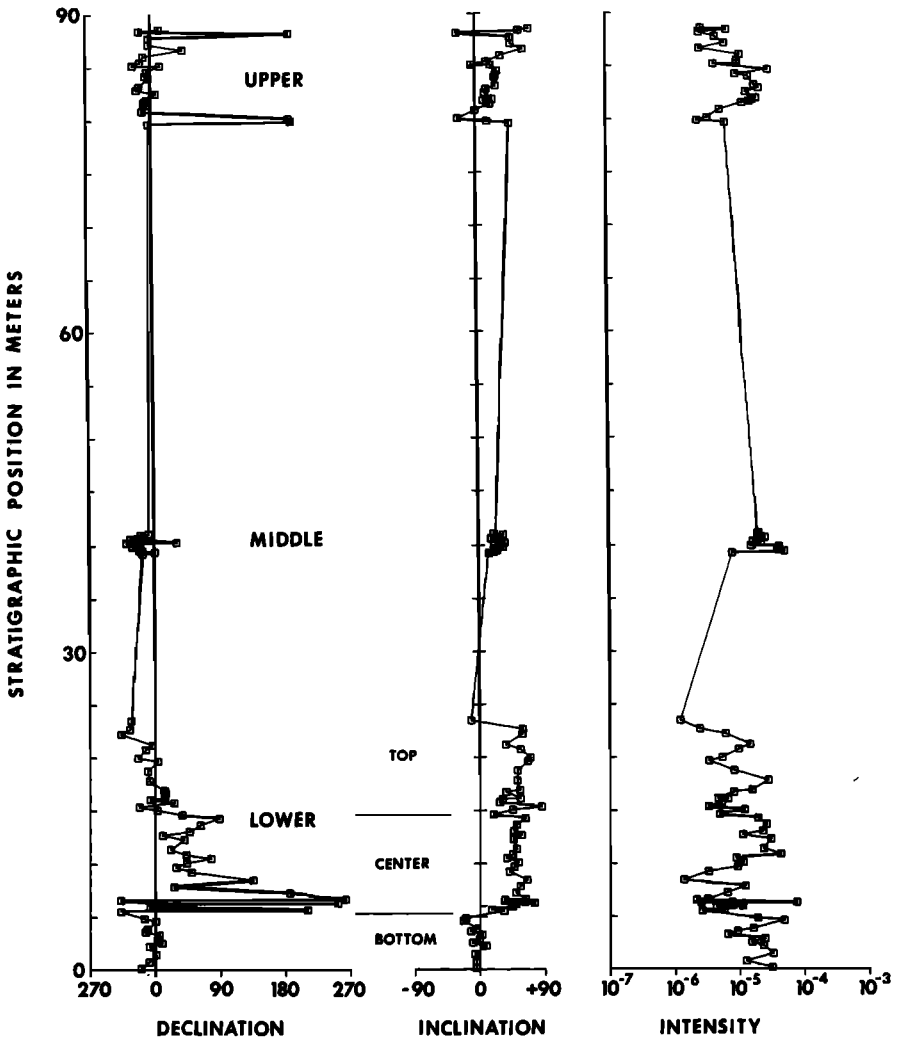


Fig. 5. Stratigraphic plot of samples from the three sampled sequences in the Kayenta formation after 630°C demagnetization. Note the three distinct directions within the lower sequence (bottom, center, and top).

pole position calculated (Table 1) includes all samples.

In the lower sequence, the mean direction showed no systematic change on heating to either 550°C or 630°C (see Figure 3). The mean direction of the lower sequence is not the same as the mean directions of the middle and upper sequences, nor is it the same as the Upper Triassic direction, which the middle and upper sequences approximate. Furthermore, there is a distinct difference in the magnetization directions in the lower sequence, with change in stratigraphic position, which became more apparent with increasing demagnetization. Three distinct groups of directions are apparent. As Figure 5 shows, moving upward in the stratigraphic sequence, the mean direction changes rather abruptly from 354.2° declination, −10.6° inclination (bottom) to 46.1°, 50.1° (center) and then to 356.2°, 47.0° (top). Stereographic plots of these three distinct groups are shown in Figure 6. Within each group the cluster of directions was better after 630°C compared with 550°C, although the dispersion is greater than

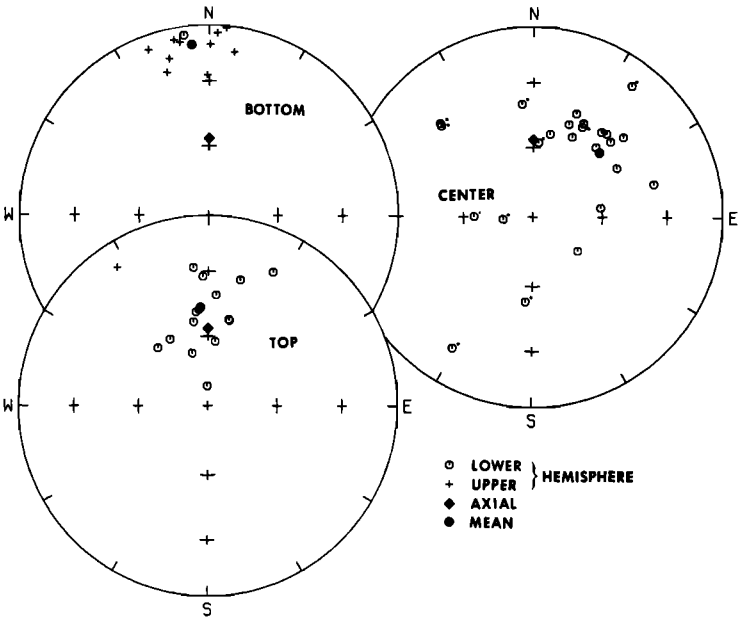


Fig. 6. Directions of magnetization after 630°C demagnetization for the lower sequence of the Kayenta formation with the bottom, center, and top intervals shown in Figure 5 plotted separately.

it was in the NRM direction. Mean directions and poles were computed for each of the three groups after removal of the scattered section between the bottom and center portions (see Figure 5) and the single transitional sample between the center and top portions (see Table 1).

The individual pole positions calculated for the upper two sequences in the Kayenta formation and for each of the three sections in the lower sequence are shown in Figure 7 against a background of all pole positions from demagnetized data published thus far for the North American Upper Triassic [*McElhinney*, 1968*a*, 1968*b*, 1969; *Irving*, 1964]. The middle- and upper-sequence poles are very similar and are insignificantly different from the published Upper Triassic pole positions. The lower-sequence poles are distinctly removed from the group of Upper Triassic poles. The stratigraphically lowest and highest directions in the lower sequence (bottom and top) gave poles that lie on the edges of the entire Triassic population of pole positions, whereas the center-direction pole obtained lies far from other Triassic poles. Since the significance of the three directions of the lower sequence is not clear, a single pole position was not calculated for these data, and the lower sequence is ignored in future references to the Kayenta pole positions. The α_{95} ovals of confidence of the middle- and upper-sequence poles overlap each other, and their agreement with the group of published Upper Triassic poles indicates that they are a good estimate of the pole position for North America during uppermost Triassic-lowest Jurassic time.

CARMEL FORMATION

A marine-nonmarine transitional deposit, the Carmel formation, was sampled north of Hanksville, Utah, on the southeastern margin of the San Rafael swell, west of Utah Highway 24 (38.6°N, 110.7°W). The sampled section dips 11° SE and consists of a basal coarse sandstone overlain by red to purple shales with calcareous shales and limestone above. A stratigraphic section of 10 meters beginning at the Navajo-Carmel contact was sampled at an average interval of 20 cm (47 samples). Marine fossils in the lower limestone units of the Carmel have established a Middle and lowest Upper Jurassic age for the formation [*Imlay*, 1952; *Smith et al.*, 1963].

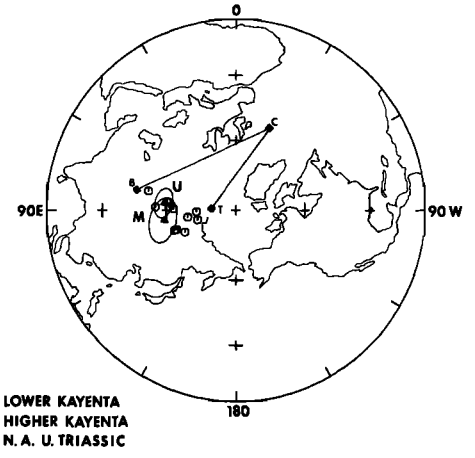


Fig. 7. Kayenta pole positions and all published demagnetized Upper Triassic poles from North America. The upper and middle sequences are shown with their oval of confidence. Poles of the three separate directions of the lower sequence (bottom, center, and top) are shown by diamonds.

The NRM directions from the Carmel formation were aligned along the direction of the axial geocentric dipole of the earth's magnetic field before structural correction (Figure 8*a*). The mean of the NRM directions was 357.6° declination, 50.5° inclination. The samples had a mean NRM intensity of 5×10^{-6} emu/cm^3, approximately the same as the Kayenta samples and other stable red sediments that have been studied paleomagnetically [*Helsley*, 1969; *Picard*, 1964].

Pilot samples were demagnetized in steps of 50° or 100°C, beginning at 150°C and continuing to 650°C. The samples generally maintained their NRM directions until they were heated to 500°C. At temperatures between 550°C and 600°C, directions of some samples began to move by large amounts, but they never regrouped, even when heated to 650°C (Figure 8*c*).

All samples were heated to 500°C and 550°C. The majority (34 out of 43 samples) stayed near the direction of the axial dipole field or moved to slightly shallower inclinations (Figure 8*b*). This behavior was unexpected, since the lithology of the Carmel formation has an appearance identical to the lithology of other red sediments that have produced reliable paleomagnetic results. Apparently characteristics of dark red color and fine grain size are not always reliable indicators of magnetic stability. From the few samples that moved significantly on demagnetization, one is given the impression that the Carmel formation may have initially possessed a reversed Jurassic (?) magnetization and has been very strongly, almost completely, remagnetized sometime during the Cenozoic.

Entrada Formation

The massive buff to light red or orange earthy sandstones and siltstones of the Entrada formation [*Wright and Dickey*, 1958] were sampled in two localities. The main locality in strata dipping 23° ESE is located along the Muddy River (110.9°W, 38.5°N) just east of the San Rafael swell. Another locality in flat-lying beds was sampled 25 km to the southeast of this locality, along Utah State Highway 24, at the base of a small butte at 110.7°W, 38.5°N.

The NRM directions of the samples from both localities were the same as the direction of the present field before structural correction (see Figure 8*d* and 8*e*). The mean of the untilted samples, 12.2° declination, 60.1° inclination, was identical to the mean from the tilted strata before correction for the dip of the beds, 8.5° declination, 60.3° inclination. Their NRM intensities were almost an order of magnitude higher (2×10^{-5} emu/cm^3) than intensities in other red sediments, probably indicating a large addition of secondary magnetization parallel to the present field direction to whatever original magnetization may have been present.

Nine samples heated in 100° steps from 200°C to 550°C showed little change from the NRM directions for temperatures below 500°C. Heating at approximately 575°C caused large scatter in the magnetization directions (Figure 8*f*). Intensities decayed rapidly during the heatings and, after 575°C, were slightly less than two orders of magnitude lower than NRM intensities. As a result of the unsuccessful pilot-sample demagnetizations, no additional work has been done.

Summerville Formation

The Summerville is predominantly a fine-grained chocolate brown sandstone. The very thin and evenly bedded formation has been described south of the sampling area by *Smith et al.* [1963]. The Summerville is gradational

into the underlying Curtis sandstone. Fossils (ammonites) in the underlying Curtis formation establish an early Upper Jurassic (lower Oxfordian) age for the Curtis [*Gilluly and Reeside*, 1928; *Imlay*, 1952] and therefore imply an Upper Jurassic (possibly upper Oxfordian) age for the Summerville formation.

Northward of the Entrada and Carmel localities, on Utah Highway 24, the Summerville crops out in cliffs of very slightly dipping strata (5° ENE). Just west of the junction of the San Rafael River with Highway 24 (110.1°W, 39.9°N), 52 samples were collected in stratigraphic succession from 10 meters of section near the top of the Summerville formation.

The NRM directions showed large scatter over the lower hemisphere, with a large cluster of directions in or near the present direction of the magnetic field. A few samples showed reversed directions, and others suggested the addition of a vector in the direction of the axial dipole field (Figure 9*a*). NRM intensities were slightly lower than in Kayenta or Moenkopi samples, with a mean of 3×10^{-6} emu/cm^3.

Twelve pilot samples were thermally demagnetized in eight steps between 150°C and 650°C. Most sample magnetization directions changed significantly with progressive heating and most markedly after heating to 150°C. Smaller directional changes continued to occur after each heating, including the 550° and 600°C heatings. During thermal demagnetization the sample directions grouped into two polarities. On heating to temperatures between 100° and 200°C the intensities of pilot samples with reversed directions either rose noticeably or dropped and then rose, reflecting the removal of a component of varying intensity aligned approximately in the direction of the axial dipole field. Pilot samples of normal polarity showed continuous intensity decreases throughout the heatings. The largest changes in intensity for both polarities were after heating to temperatures of 150°C and 600° to 650°C, as shown in Figure 9*b* and *c*. From the information gained from

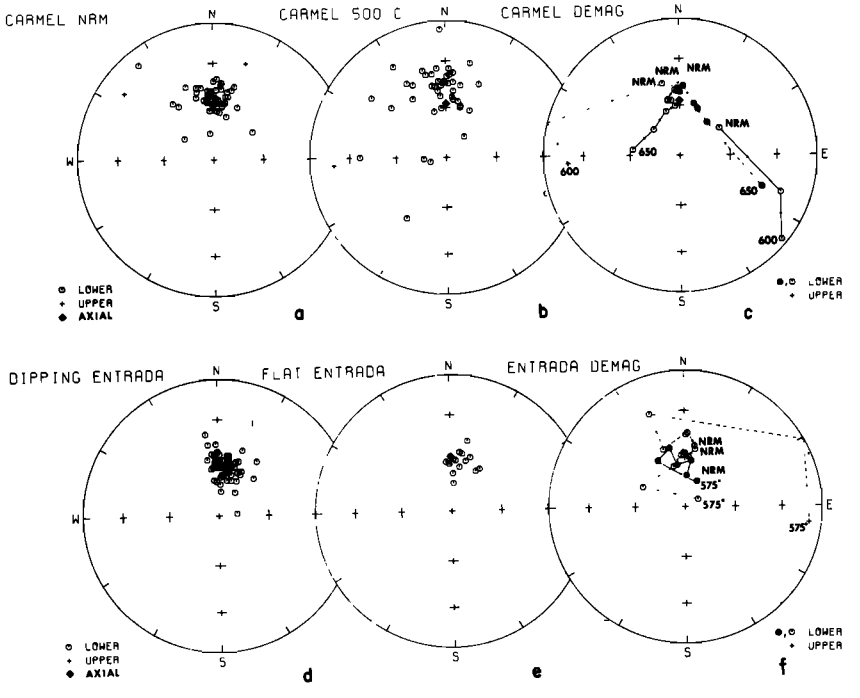


Fig. 8. Carmel and Entrada magnetization directions. Carmel directions of (*a*) NRM and (*b*) after 500°C demagnetization; (*c*) directional behavior of four Carmel samples during thermal demagnetization. Entrada NRM directions from (*d*) dipping and (*e*) flat strata without structure correction); (*f*) behavior of three samples on demagnetization to 575°C.

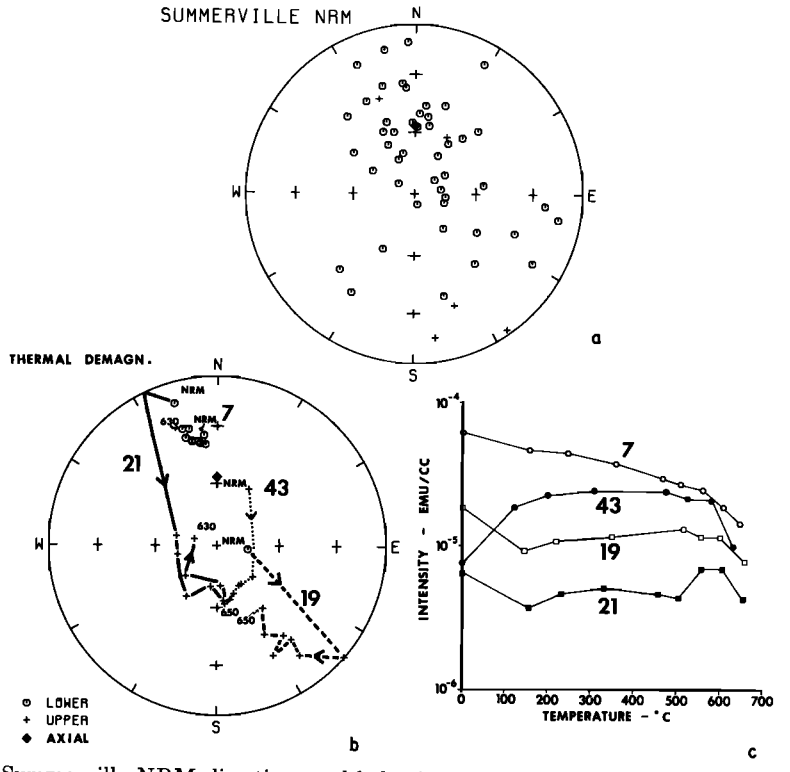


Fig. 9. Summerville NRM directions and behavior of four samples on thermal demagnetization to 630° or 650°C.

the pilot demagnetization experiments, the remaining 40 samples were demagnetized in three steps at 400°, 500°, and 550°C. They behaved in the same manner as the pilot samples. A moderately good grouping of magnetization directions was obtained by 550°C (Figure 10*a*). Intensities on demagnetization behaved normally over all in that they dropped only 1 order of magnitude or less to about 0.5×10^{-6} emu/cm^3.

The uppermost 12 samples (1.7 meters) in the stratigraphic sequence are of normal polarity, whereas the rest are of reversed polarity. The mean direction of the normal

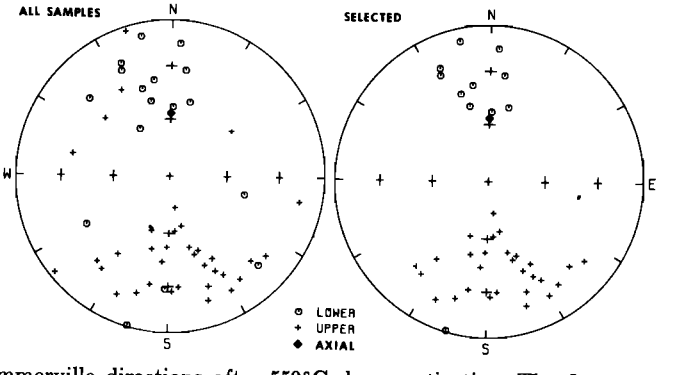


Fig. 10. Summerville directions after 550°C demagnetization. The figure on the left shows all samples; the right-hand figure shows the group after scattered samples (see Figure 11) have been removed.

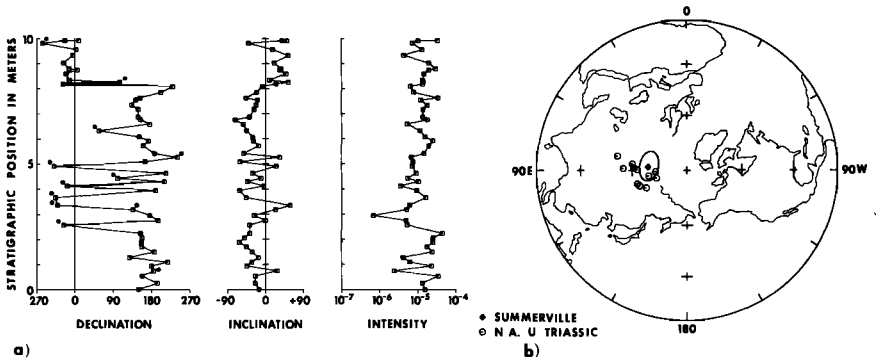


Fig. 11. Summerville directions after 550°C demagnetization plotted according to stratigraphic position (*a*). Dots near sample declination points indicate randomly scattered samples omitted in Figure 10*b*. (*b*) shows the Summerville pole position and oval of confidence with all demagnetized Upper Triassic poles from North America.

samples is 350.2° declination, 34.9° inclination; the mean direction of the reversed samples is 174.6° declination, −39.1° inclination. These two means are very close to 180° apart, even though considerable scatter is present. It was observed that the scattered directions are randomly distributed in the stratigraphic sequence. Therefore, samples that are stratigraphically part of either polarity group but have directions completely removed from the majority of the group (Figures 11*a* and 10*b*) were removed, and another analysis was done. Normal and reversed means after removal of the widely scattered data are very similar to the normal and reversed means of all the data (Table 1), seemingly indicating that a stable component of the magnetization had been isolated in these samples. The pole position calculated from the selected portion of the data is plotted in Figure 11*b*. The Summerville pole is about 10° farther north than the Kayenta poles and is more northerly than the majority of Upper Triassic poles, but it is much closer to the Triassic direction than it is to the Cretaceous direction.

Discussion of Results

The magnetization directions determined from the middle and upper parts of the Upper Triassic–Lower Jurassic Kayenta formation are the same as directions observed for Triassic time. The lower Kayenta is anomalous, and the Carmel and Entrada formations are unstable, having magnetizations aligned in the direction of the axial dipole field. Presumably directions of any original Carmel or Entrada magnetization would have been very similar to the Kayenta directions, since the Upper Jurassic Summerville formation has a mean direction not far removed from that of the Kayenta formation.

It would seem, on the basis of the stability of the Summerville data, that the Summerville formation was magnetized before the pole (or continent) had moved significantly from the Triassic and Lower Jurassic position toward the Upper Cretaceous position of the North American polar-wander curve. This pole position seems to indicate that the movement was at most only just beginning in Upper Jurassic time (Figure 12) and that the largest amount of movement occurred in the latest Jurassic time and/or the Early Cretaceous. The Summerville pole position further restricts the geologic time period during which a rapid movement of 30° arc distance may have taken place. It seems likely that the 30 degrees of movement may have occurred in about 30 m.y. This amount of time is the length of time between the 151 m.y. upper limit of the Oxfordian (Summerville) [*Geological Society of London*, 1964] and the oldest radiometrically dated poles of Cretaceous age, Mt. Ascutney, 125–135 m.y. [*Opdyke and Wensink*, 1966] and the Monterigian Hills, 100–122 m.y. [*Larochelle*, 1961]. The rate of apparent polar movement was therefore relatively rapid, about 10 cm/yr. This rate of movement of the pole, or the continent with respect to

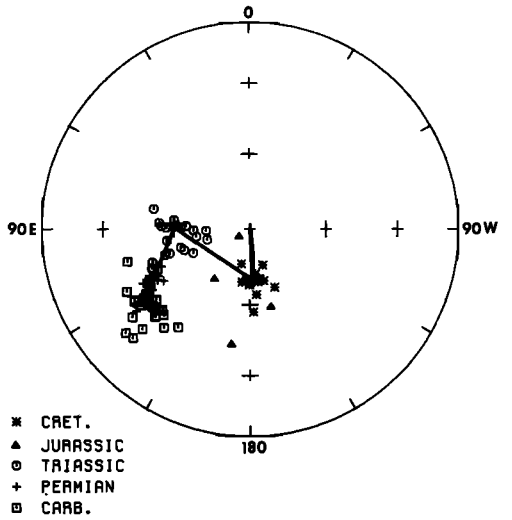


Fig. 12. Upper Paleozoic and Mesozoic pole positions from North America as in Figure 1, with the Kayenta middle and upper sequence and the Summerville pole positions.

the pole, is comparable to the spreading rates deduced from sea-floor spreading during parts of the Tertiary.

If one interprets these results in terms of continental drift, the data give information about the time of opening of part of the Atlantic Ocean. The Upper Triassic grabens of the east coast of North America suggest that the time of inception of rifting in the North Atlantic was Upper Triassic. The Atlantic may have opened slightly between Upper Triassic and Upper Jurassic, but the Summerville pole position indicates that the major amount of the separation in the central Atlantic Ocean (separation of North America from Africa) did not take place until Upper Jurassic to Lower Cretaceous times.

Magnetic anomalies in the Atlantic sea floor are tentatively dated back to the Upper Cretaceous. *Pitman et al.* [1971] have delineated the anomaly pattern as far back as anomaly 31 (72 m.y.). Beyond the Upper Cretaceous anomaly 31 a series of anomalies and 'quiet zones' are present that account for approximately 54% of the central Atlantic. If major movement did not occur until Upper Jurassic, sea-floor spreading must have been exceedingly rapid during the Upper Jurassic and Lower Cretaceous to produce 54% of the ocean basin by Upper Cretaceous time.

The Jurassic age of the sea floor has been shown from Joides drilling. Upper Jurassic (Oxfordian, Cavollian?) limestone and clay were recovered from the western margin of the Atlantic [*Ewing et al.*, 1970; *Laughton et al.*, 1970]. Upper Jurassic aptychus limestones containing radiolaria are found in the Cape Verde Islands on the eastern margin of the Atlantic [*Furon*, 1963; *Colom*, 1955]. The Summerville pole is not in conflict with these data, since a small amount of opening (10°–15°) would give a deep, narrow seaway, permitting these deposits to form, but would go unnoticed in the uncertainties inherent in the paleomagnetic data.

It seems reasonable to relate the two abrupt changes of direction in the North American polar-wander curve to changes in the pattern of drift between North America and both Europe and Africa. Thus the drift of North America away from Africa would produce the apparent movement of the pole with respect to the North American continent from the Triassic through Upper Jurassic position on the polar-wander curve (approximately 90°E, 60°N) to the Cretaceous position (about 180°, 70°N). The continued drift of North America relative to both Europe and Asia would then be reflected by the Upper Cretaceous and Tertiary apparent movement of the pole from 70°N to 90°N. The presence of sea floor extending beyond the Cretaceous anomaly 31 in the central Atlantic (between Africa and North America) and the absence of such sea floor between Europe and North America supports the two-stage opening of the North Atlantic as deduced from the North American polar-wander curve.

Conclusion

In summary, the paleomagnetic data from the Kayenta and Summerville formations indicate that the major movement of North America with respect to the pole did not begin until Upper Jurassic (post-Summerville) time. Moreover, the paleomagnetic data for North America indicate that two distinct episodes of drift have taken place, one in Upper Jurassic to Cretaceous times and one in the Upper Cretaceous and Tertiary. These are reflected in the opening of the central Atlantic and the North and central Atlantic, respectively, and are com-

pletely compatible with and reinforced by the sea-floor spreading data.

Comparison of the sea-floor spreading data, the paleomagnetic record, and the geology of the Atlantic basin as deduced from the Joides drilling all suggest that rifting is a two-stage process involving early initial rifting with little spreading, perhaps extending over a considerable period of time (Upper Triassic to Upper Jurassic in the case of the Atlantic), followed by a period of rapid spreading in which the major portion of the ocean basin is formed in a relatively short period of time. The East African rift valley and the Red Sea may be more recent examples of the first stage of this two-stage process.

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